



Session 2: Ecotoxicological Databases

Theoretical Ecotoxicology

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ESSEN**

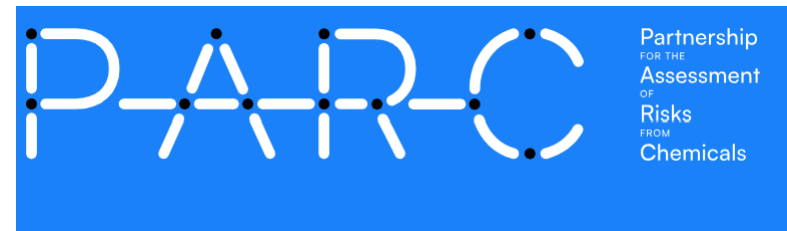
Offen im Denken

About me

- **Paul Ozoh** (Master in Ecotoxicology)
 - Doctoral Candidate

- **Research interests:**
 - Large-scale ecotoxicological risk analysis
 - Chemical risk assessment
 - Data analytics and visualisation

- **Current work:**
 - “**Benchmarking pesticides** environmental risk assessment to support a transition towards a holistic paradigm”

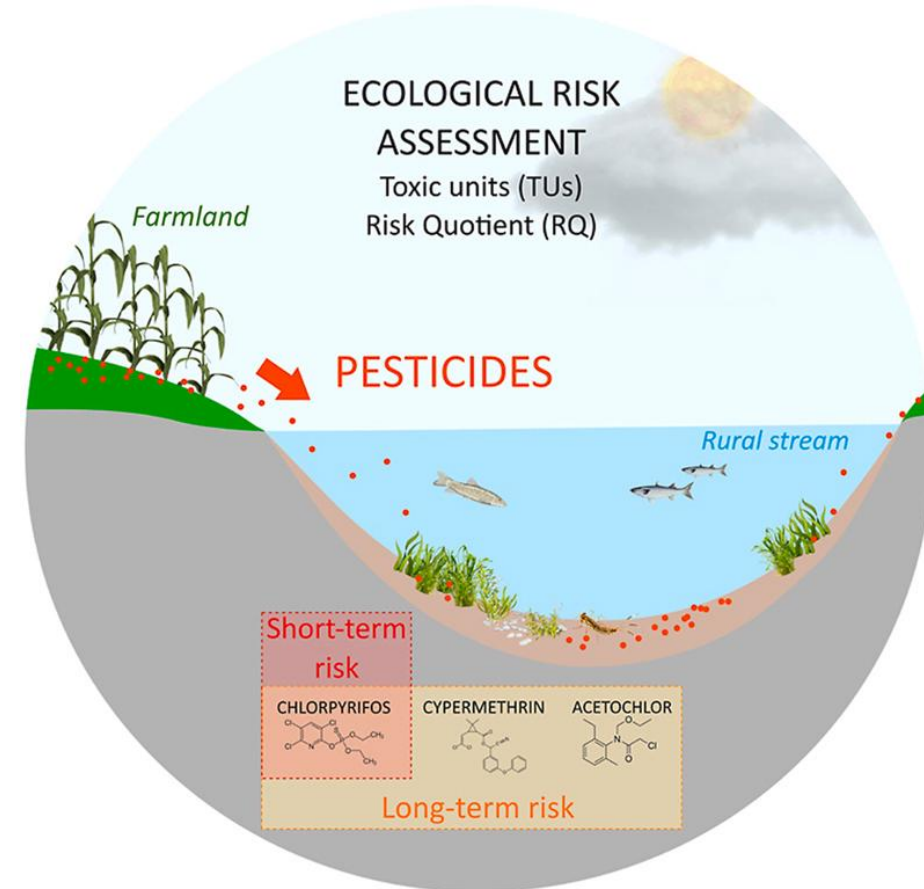


Learning objective

- ❑ Understand the ERA concept and data sources
- ❑ Extract endpoints from databases
- ❑ Use automated workflows
 - Automate data retrieval and processing (R)
 - Perform basic pesticide risk assessment metrics (RQ)
 - Build a mini reproducible workflow

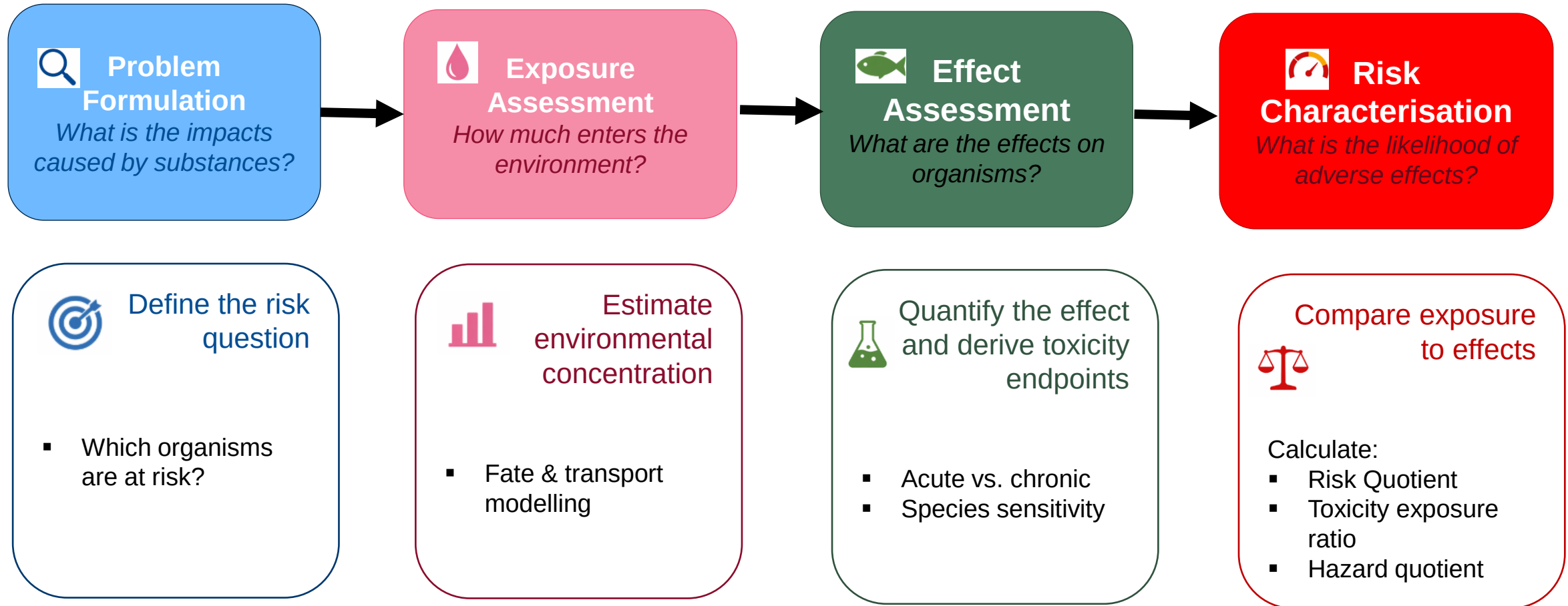


Ecological Risk Assessment (ERA)



Source: <https://doi.org/10.1016/j.chemosphere.2022.137598>

From hazard identification to risk characterisation



Risk Concept (Aquatic)

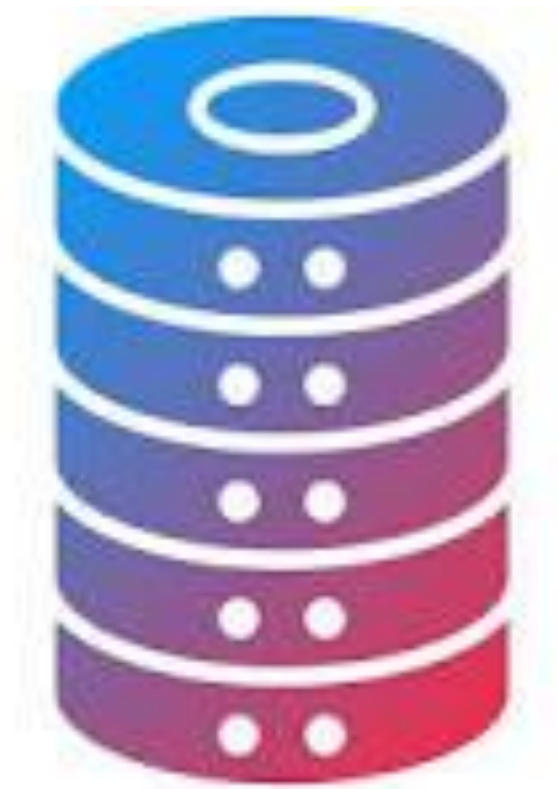
- ❑ **Effect** ► Toxicity endpoints: Acute (e.g., LC_{50} , EC_{50}), Chronic (e.g., NOEC)
- ❑ **Exposure** ► Predicted Environmental Concentration (PEC)
- ❑ **Assessment (safety) factors (AF)** ► between 3 - 1000; applied to toxicity endpoints (e.g., $LC_{50} / 100$)
- ❑ **Risk Quotient (RQ):**
 - $RQ = PEC / RAC$
 - $RQ < 1$ acceptable
 - $RQ > 1$ risk

Regulatory Acceptable Concentration (RAC) = Toxicity endpoint / AF

➡ Output: point estimate of risk



Databases



EU Pesticides Database

- A centralised, official online resource maintained by the European Commission.

Structure :

- **Active substances:** Lists approved, non-approved, or candidate substances with toxicological reference values.
- **Maximum residue levels (MRLs)**
- **Safeners, synergists, and adjuvants:**



<https://ec.europa.eu/food/plant/pesticides/eu-pesticides-database/start/screen/active-substances>

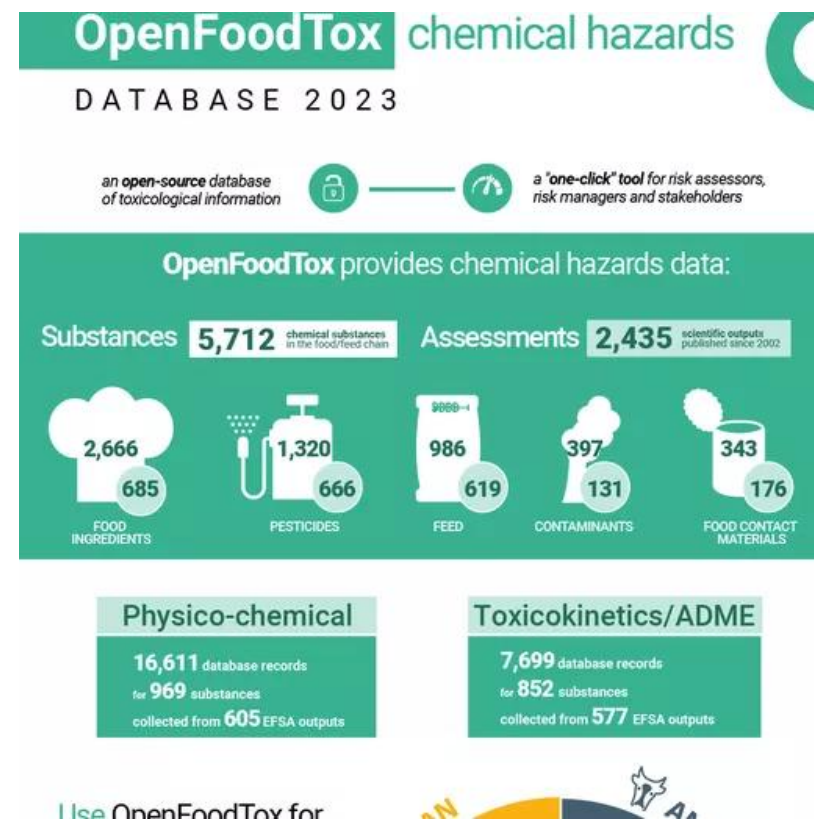
Chemical Hazard Information Databases

EFSA OpenFoodTox

- OpenFoodTox 2.0 integrates over 11,000 toxicity studies and 16,000+ physicochemical records.

Structure:

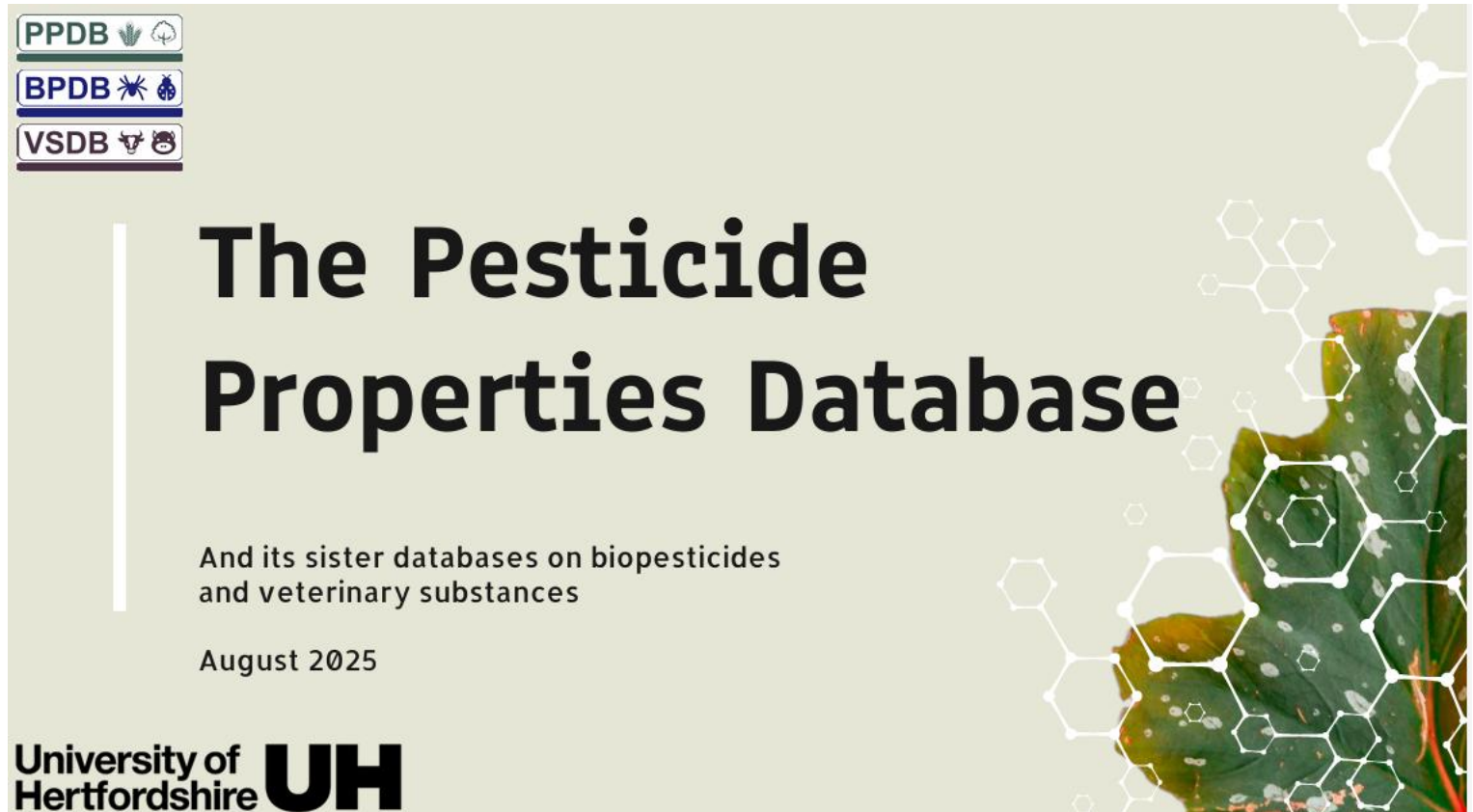
- Substance characterisation and EFSA outputs
- Reference points and reference values
- Genotoxicity results
- Pharmacokinetic and toxicokinetic data
- Files are downloadable in Excel or CSV format via EFSA's Scientific Data Warehouse and Zenodo.



<https://www.efsa.europa.eu/en/microstrategy/openfoodtox>

PPDB Overview

A relational database with identity, physico-chemical, fate, ecotox, and use data for many pesticides, designed to support risk activities.



The graphic features three stacked logos in the top left: PPDB (with plant and leaf icons), BPDB (with insect icons), and VSDB (with animal icons). The main title 'The Pesticide Properties Database' is in large, bold, black font. Below it, in smaller text, is 'And its sister databases on biopesticides and veterinary substances'. The date 'August 2025' is centered below that. The University of Hertfordshire logo (UH) is in the bottom left. The background is a light green with a molecular structure overlay on the right side, which includes a close-up of a green leaf with white spots.

PPDB

BPDB

VSDB

The Pesticide Properties Database

And its sister databases on biopesticides and veterinary substances

August 2025

University of Hertfordshire **UH**

<https://sitem.herts.ac.uk/aeru/ppdb/>

Why automation?

Challenges:

- Many substances, reproducibility, transparency, updating

Advantage of scripted access:

- Documented queries
- Easy re-run if database updates
- Direct link from data retrieval to R-based analysis

Automation Pipeline



Example

- Calculation of aquatic organism **risk quotients** (RQ):
Exposure / Toxicity (incl. AF) > 1 → **Risk**
Exposure / Toxicity (incl. AF) < 1 → **No Risk**
- Example:
 - Exposure = PEC = 0.5 mg/L
 - Toxicity = EC₅₀ = 1 mg/L
 - AF = 100 (Acute Aquatic organism –Tier 1)

$$RQ = \frac{PEC}{(Toxicity/AF)}$$

$$RQ = 0.5/(1/100) = 50 \rightarrow \text{Risk or No Risk?}$$



Further Reading

1. <https://www.efsa.europa.eu/en/applications/pesticides>
2. [https://sitem.herts.ac.uk/aeru/vsdb/docs/PPDB_presentation.](https://sitem.herts.ac.uk/aeru/vsdb/docs/PPDB_presentation)
3. <https://pubmed.ncbi.nlm.nih.gov/39181462/>
4. <https://pepijn-devries.r-universe.dev/ECOTOXr>



Exercise